

# Designing Digital Content to Accommodate for Colour Vision Deficiency

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## ABSTRACT

Digital content and designs feature coloured components in an effort to provide information, however, this information can be lost for those who have Colour Vision Deficiency (CVD). Over the past three decades research to provide digital aids has progressed by creating both tools that enable for simulations of types and severities of CVD and various corrections through recolouring. However, simulations have often been overvalued and in many cases fully replace CVD lived experiences in design, and recolouring tools tend to take a destructive approach modifying all colours in an attempt to ‘correct’ for CVD. This research first looks to understand the dangers in relying on simulations for both designs and assistive tools. Next, I look to understand the current perspectives and use cases of current assistive technologies like recolouring by those with CVD. Finally, I develop guidelines to show how design can be done to better accommodate those with CVD in the design of coloured interfaces.

## CCS CONCEPTS

• Human-centered computing → Accessibility.

## KEYWORDS

Colour Vision Deficiency, Recolouring, Colour Vision Deficiency Simulation, Colour Patterns, Inclusive Design

### ACM Reference Format:

Connor Geddes. 2023. Designing Digital Content to Accommodate for Colour Vision Deficiency. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3544549.3577047>

## 1 INTRODUCTION

Designing to assist those with Colour Vision Deficiency (CVD) has taken a backseat to research focused on ‘curing’ or ‘correcting’ CVD. Over the past 30 years, both tools & algorithms that simulate varying types and severities of CVD (simulations) and at the same time algorithms that manipulate the screen colours (recolouring

algorithms) have been rapidly developed and relied on. These solutions have in many cases become the de facto solution and design advice for assisting and including those with CVD.

However, simulations simplify the challenges those with CVD have in their lives and avoid addressing actual problems by understanding the actual issues those with CVD have in software or designs that make use of coloured components. Recolouring algorithms remap colours in the reduced colour perception of the individual with CVD to hopefully make colour distinction easier. However, by doing so they inherently change the way those with CVD view the content and once again (similar to simulation use) miss the point completely in understanding the actual challenges those with CVD have.

To address the choice that modern designs and solutions for assisting those with CVD have steered towards simulation use in lieu of actual people with CVD and that solutions have become focused on ‘curing’ CVD rather than accommodating, I will first focus on understanding the issues that can come from relying on simulations in designing and developing interfaces for those with CVD. Next, I will look at the limitations of recolouring tools both from the perspectives of those with CVD from their personal experiences and typical uses, and through evaluation of specific colour related tasks. Finally, I will put forth guidelines that will provide design advice for accommodating those with CVD in a wide range of potential use cases (e.g., games).

My research will look to assess the current state of accessibility practices for those with CVD. While providing advice for future paths forward that can hopefully provide guidance to designers and developers on the types of accommodation that best assist those with CVD in the digital designs they are developing.

## 2 BACKGROUND AND RELATED WORK

Colour Vision Deficiency (CVD) affects ~5% of the population in some form. The most common type of CVD is red-green CVD affecting an individual’s ability to accurately discriminate and name colours along the red-green axis. Examples of common confusion colours include ‘bright-greens’ and ‘yellows’, ‘purples’ and ‘blues’, ‘reds’ and ‘browns’, and ‘greens’ and ‘browns’. Figure 1 shows a red-green CVD simulation of red (male) and brown (female) cardinals lined up on a fence to illustrate how these colours can be hard to distinguish. This inability to discriminate specific colours can result in missing important information passed in an interface. Further an inability to correctly name colours can result in colour-communication difficulties between those with CVD and those without.

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CHI EA '23, April 23–28, 2023, Hamburg, Germany  
© 2023 Copyright held by the owner/author(s).  
ACM ISBN 978-1-4503-9422-2/23/04.  
<https://doi.org/10.1145/3544549.3577047>



**Figure 1: Images illustrating Red-Green CVD via the use of a simulation. Image on the left shows red (male) and female (brown) cardinals lined on a fence and the image on the right is a duplicate of the original cardinal image just simulated in a form of red-green CVD, showing how the red and brown cardinals become hard to distinguish**

In order to understand the issues that those with CVD could have in discriminating or interpreting colours, simulations are often employed. These approaches can help designers and developers understand colour choices that could be difficult to interpret by those with CVD. However, it should be noted that CVD simulations cannot adequately allow a non-CVD individual to successfully behave or act like someone who has CVD [9]. Simulation methods have been created based off of adjusting colours based on colours that are confused [16], on the subjective perceptions of unilateral dichromats (CVD in one eye) [3], based on the idea of the two stage theory of human colour vision [15], and based on adjusting colours to specific individuals severity of CVD based on a calibration process [8].

In order to accommodate users with CVD, recolouring tools [6, 7, 12–14, 16, 18, 21] or filters have been created. These recolouring tools will often shift or remap hues in order to improve the distinction of confusing colours for individuals with CVD. These algorithms have become so popular that they are available in the settings of Android [11], iOS [2], MacOS [1], and Windows devices [17]. Further, they are also available for addition by default in the Unreal Engine [20] allowing developers to add these filter easily to their games.

With this seeming large availability of recolouring tools and 30 years of research [16] investigating ways of developing better recolouring tools, they must be the best tool for assisting people with CVD. However, what has lacked in over 30 years of research is an understanding in how these tools are used or perceived by people with CVD. In the vast amount of research exploring recolouring tools it is always taken as a fact that they must be used in a pervasive way in which they are turned on to ‘correct’ the vision of a CVD individual when using a digital device.

Also the way recolouring tools are evaluated tends to focus on solving diagnostic tests like the Ishihara plate test [4]. Showing again this focus on attempting to ‘correct’ rather than accommodate. Further, recent research [19] has shown that of 101 CVD papers they reviewed only 57 actually conducted research with CVD participants and of those only 29 had a participant count greater than 10.

I believe this highlights important issues that need to be covered:

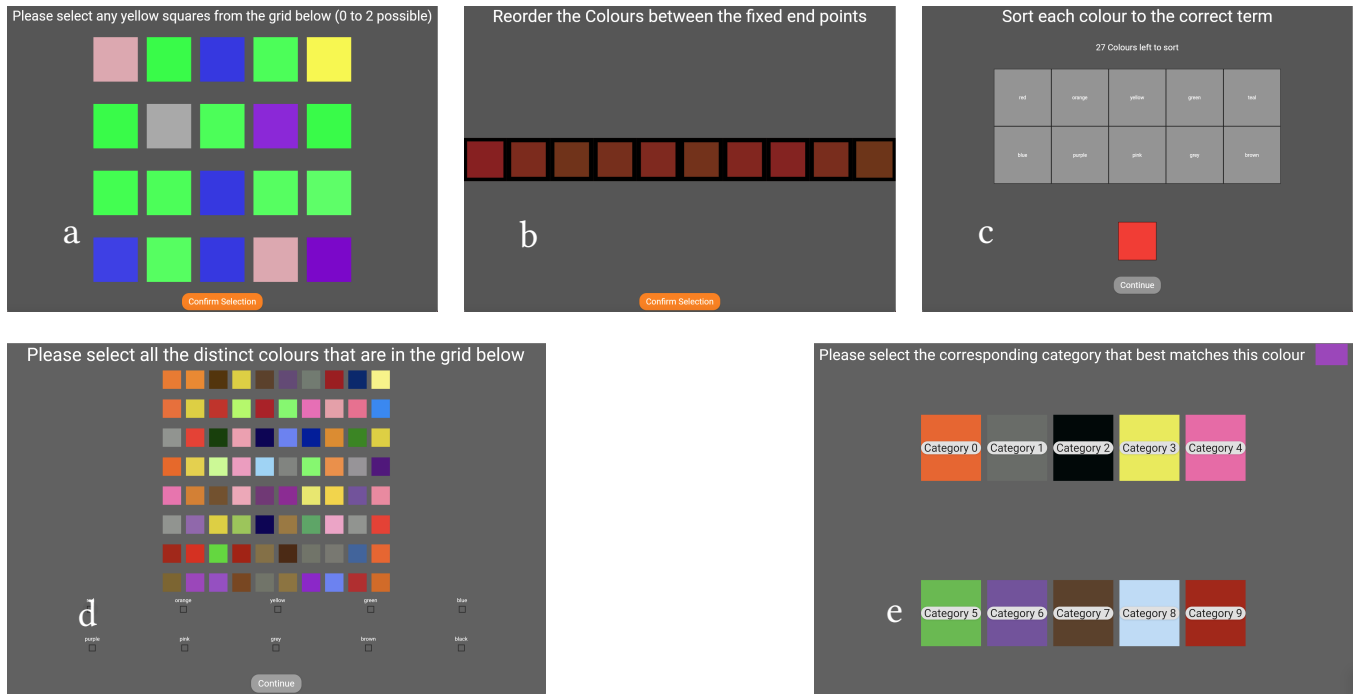
- Create tasks that better cover realistic challenges so that research can go away from attempting to solve diagnostic tests
- Explore simulations ability to convey the difficulties that those with CVD have with colours
- Understand the perspectives those with CVD have with recolouring tools
- Explore contexts (e.g., games) where colours are often importantly used to convey information and understand what assistance those with CVD want

### 3 RESEARCH OBJECTIVES

With my research I hope to clarify how digital content should be designed for people with Colour Vision Deficiency (CVD), and try to encourage that designs in the future focus on good user-centered design.

In order to accomplish these goals, the main steps for my PhD thesis will be the following:

- (1) Due to current evaluations of tools to assist those with CVD often using unrealistic tasks, I will create tasks that better cover the challenges those with CVD have with colours
- (2) As recolouring is the most common tool available for people with CVD, I want to understand how people with CVD use and perceive this tool
- (3) Next I will conduct studies with designers to understand the potential dangers simulations could pose if used in design and whether they truly can provide designers with a good understanding of the issues those with CVD have in an interface
- (4) Finally, I will look at specific cases via which colour can be applied such as games and understand the issues those with CVD have in those specific cases, and how current accommodation efforts assist them.



**Figure 2: Images showing the colour tasks developed. With (a) showing the selection task, (b) showing the transition task, (c) showing the sorting task, (d) showing the distinction task, and (e) showing the category task.**

## 4 CURRENT RESEARCH

Currently I have completed steps (1) and (2) in my research objectives. Step (1) consisted of two papers with the first presented at CHI 2022 [10] and the second at ASSETS 2022 [9]. For Step (2) this work is currently under review at CHI 2023 and has received positive feedback in the first round of reviews being recommended for revise and resubmit. Step (3) has yet to be started but I hope to start this research in the spring of 2023. Step (4) will look at both games and visualizations in two separate studies, I am starting by looking at the aspects of accommodating those with CVD in games at the start of 2023. Finally, I hope to look directly in the space of visualizations at the end of 2023 or at the start of 2024.

### 4.1 Evaluation of Colour Identification Aids

Currently to evaluate the effectiveness of a given colour identification aid, diagnostic tests like the Ishihara plate test are most commonly used. A large reason for the prevalence in evaluations stems from the ease with which these tests can be evaluated using simulations. Therefore this led to two main issues that we explored in this research: 1) improving colour tasks for the evaluation of CVD aids, and 2) Understanding whether simulations can adequately stand in for CVD participants.

A major problem in utilizing evaluations such as the Ishihara plate test is it has a large reliance on comparative colour judgement. So to address these challenges, I created five colour identification tasks to better cover the four types of colour challenges specified by Cole [5]: (1) comparative: where colours must be compared to find a difference, (2) denotative: where colours are referenced by

name, (3) connotative: where colour has some explicit or assigned meaning, and (4) aesthetic: where colour is used to convey beauty.

The first task created was the selection task and this task required participants to identify colour by name (denotative) with the assistance of comparison (Figure 2 (a) shows the selection task). The second task was the transition task and this required participants to order eight randomly assorted colour swatches between two anchor colours, when ordered they would form a hue scale (Figure 2 (b) shows the transition task).

The third task was the sorting task and this required participants to sort colours swatches into the correct corresponding colour name bucket (Figure 2 (c) shows the sorting task), allowing participants to only see one colour swatch at a time (completely denotative). The fourth task was the distinction task and this required participants to determine the number of unique hues present from a listing of hue categories presented in check boxes below (Figure 2 (d) shows the distinction task). Finally, the fifth task was the category task and it required participants to identify the correct category based on a target category presented (Figure 2 (e) shows the category task).

We found that these tasks were challenging for our participants with CVD, and at the same time we saw that the non-CVD participants were able to complete the tasks with little challenge. This shows some potential for the adaptation of our tasks in future studies looking to accommodate those with CVD.

Next, to explore whether simulations could indeed substitute for an individual with CVD, we set out to compare non-CVD participants using simulations of various severities to CVD participants.



**Figure 3: Images illustrating how recolouring algorithms work. The left image shows how the red cardinals are remapped to pink and the right image simulates red-green CVD on that same image showing how this distinction should be easier.**

We found differences both in accuracy and completion time when comparing the results of simulated non-CVD participants to CVD participants in all tasks. We attribute the difference and often large difference in completion time on many tasks (Selection, Sorting, and Category) to be the result of those participants with CVD using a life-time of knowledge knowing how colours are mapped under their vision to narrow down colours more selectively and more rigorously, compared to those without CVD who when under a simulation showed behaviour that didn't differ too much from our non-CVD control group.

## 4.2 Recolouring Tool Use

Despite over 30 years of research looking at recolouring tool use, I found there has yet to be any research understanding how people with CVD apply them to solve colour related tasks. Furthermore, in general there was no research understanding the perceptions those with CVD have on recolouring. In order to address this current gap I conducted a combination of a scrape of the r/colorblind subreddit and an observation study with follow on interview.

Our findings indicated that those with CVD had often negative perspectives towards recolouring filters often believing that designers that implemented them didn't understand the problems they actually had with their software or were applying a quick fix to appease those with CVD. Most recommendations we saw and heard suggested that designers and developers should attempt to understand the actual problem in software not apply a blanket filter in 'guessed' hopes that it will solve all problems.

Further, We also saw that participants very rarely utilized the filter when performing colour-related tasks in an always on setting. More accurately we saw that participants would turn on and off the filter, and switch between the filters made for different types of CVD. When doing this, participants were trying to take as much information from the different states to try to see what happened to the colour and how it transformed. Participants would attempt to use this information to make judgements on what the colour name was or if indeed the colour was different. We believe the way participants used recolouring tools shows a whole new area for researchers to explore understanding potential interaction techniques to be applied to tools to best assist those with CVD.

## 5 EXPECTED FUTURE WORK AND CONTRIBUTIONS

The next steps for my research will be to first look into how simulation use can affect how designers accommodate those with CVD, I suspect that if relying on simulations designers can find issues that 'could' be a problem for those with CVD. But at the same time I would suspect that investigating design issues in this way likely pushes designers into thinking of these design issues as a colour issues and not a fundamental issue of communicating information. I believe that this issue can best be explored by conducting a workshop with designers and attempting to design several user interfaces. I believe splitting the designers into two groups one attempting to identify and address issues only with simulations and the other with the assistance of CVD participants. I believe this will produce important new information from both understanding how designers interpret the use of the simulations on their designs and the differences between the final results of the two groups.

After that, I will investigate current efforts for inclusion in the domains of games and visualizations. I chose both of these subfields as they use colours in different ways in order to convey important information to the user. After collecting all available standards in both fields I will conduct surveys and interviews in order to understand the perspectives of those with CVD on how they believe they can best be assisted. As I learned in my study looking at recolouring filters, those with CVD have a lot of knowledge on the issues that affect them and will have the best information on how they can and want to be assisted.

My research will contribute to pushing design and research for accommodating people with CVD away from 'cures'. Further, it looks to largely turn focus away from tools such as recolouring filters and CVD simulations. I believe that recolouring is a very idealized solution for accommodating people with CVD, but a more realistic solution for accommodating those with CVD in software is more nuanced and requires a direct understanding from those with CVD in specific. Going further, while CVD simulations can give an idea of the issues those with CVD have, I suspect they will tunnel vision designers into accommodating based on colour and missing more nuanced designs that could better accommodate those with CVD in interface designs.

## 6 ACADEMIC STATUS

I am pursuing a doctorate degree in computer science under the supervision of my advisor David Flatla. I started my PhD in October 2021 at the University of Guelph, I recently started my second year. I anticipate I will be done the degree within the standard length in Canada of four years, in October 2025.

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